Masters Thesis

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Abstract

Machine learning-assisted prioritization has substantially reduced the workload of systematic review screening, but determining when it is safe to stop screening remains an open methodological problem. This thesis develops and evaluates the first Bayesian stopping rule for machine learning-assisted systematic review screening. The screening process is modeled as a biased urn process in which both the total number of relevant documents and the ranking bias parameter are treated as unknown quantities to be inferred jointly from the observed screening sequence. Stopping is triggered once the posterior probability of having achieved a target recall level exceeds a user-specified confidence threshold. The method is evaluated through Monte Carlo simulations across four experimental configurations and applied to real-world screening data from ten completed systematic reviews. Results show that the Bayesian method achieves consistently lower miss rates than a frequentist baseline at every confidence level tested, at the cost of substantially lower work savings. A robustness analysis under time-varying bias reveals that the method is sensitive to violations of its constant bias assumption in a structured and predictable way, and that a simple conservative adjustment to the likelihood extends the range of assumption violations under which the recall guarantee is maintained. To the best of the author's knowledge, this is the first stopping criterion for systematic review screening grounded in a fully Bayesian framework.

1 Introduction

Systematic reviews occupy a central role in evidence-based research by synthesizing findings from large bodies of scientific literature. However, the identification of relevant studies for inclusion in a systematic review typically requires screening thousands of titles and abstracts, making the process both time-consuming and resource-intensive (O'Mara-Eves et al., 2015). The problem is compounded by the continuous growth of the published literature: the volume of scientific output has expanded to the point where keeping pace with new evidence through manual review alone is increasingly impractical (Bastian et al., 2010). In response, machine learning-assisted screening methods have been proposed to accelerate this process by ranking documents according to

their predicted relevance, allowing reviewers to prioritize screening documents that are more likely to be relevant (Cormack and Grossman, 2014).

While machine learning ranking can substantially reduce the workload associated with systematic review screening, an important methodological question remains: when is it safe to stop screening? Stopping too early risks missing relevant studies and introducing bias into the resulting evidence synthesis (Callaghan et al., 2024), while screening all documents removes the efficiency gains offered by machine learning methods. The absence of a trusted stopping criterion has concrete consequences for practice: the Cochrane Handbook for Systematic Reviews of Interventions — the authoritative methodological guide for evidence synthesis — does not currently recommend any stopping criterion for machine learning-assisted screening, precisely because no existing method can provide a true statistical guarantee that a sufficient proportion of relevant studies has been identified (Higgins et al., 2019). This problem is not unique to systematic reviews. In legal discovery, technology-assisted review must identify all disclosable documents; in hate speech moderation and fraud detection, ranked document collections must be screened until reviewers can be confident that substantially all relevant instances have been found. In each of these contexts, the need for a stopping criterion that provides a genuine statistical guarantee is both practically important and methodologically unresolved. To address this challenge, several approaches have been proposed that aim to estimate the proportion of relevant documents already identified during screening, commonly referred to as recall. In practice, systematic review guidelines typically require a high recall threshold (for example, 95%) to ensure that the majority of relevant studies have been identified (Higgins et al., 2019).

A promising approach to this problem is the use of Bayesian stopping rules, which provide a probabilistic guarantee that recall has exceeded a predefined target threshold. In this framework, the screening process is modeled as a sequential sampling problem in which documents are screened one at a time. As screening progresses, Bayesian inference is used to update the posterior distribution over unknown quantities: the total number of relevant documents and the bias introduced by the ranking model (Gelman et al., 2013). Screening can then be stopped once the posterior probability that recall exceeds a target threshold becomes sufficiently large (Callaghan and Müller-Hansen, 2020).

In this thesis, a Bayesian stopping criterion for machine learning-assisted systematic review screening is implemented and evaluated using simulation experiments. The screening process is modeled as a biased urn process, in which relevant documents are more likely to appear earlier in the screening sequence due to the ranking produced by a machine learning classifier (Callaghan and Müller-Hansen, 2024a). The degree of ranking effectiveness is captured by a bias parameter that determines how strongly relevant documents are prioritized during screening. A Bayesian inference procedure is used to sequentially update the posterior distribution over the total number of relevant documents and the bias parameter based on the observed screening outcomes (Gelman et al., 2013).

To evaluate the performance of the proposed stopping rule, a series of Monte Carlo simulation experiments are conducted. These simulations generate syn-

thetic screening sequences under controlled conditions and measure the resulting recall at the stopping point as well as the proportion of screening effort saved (Repke et al., 2026). Particular attention is given to the lower tail of the recall distribution to assess whether the stopping rule successfully satisfies the desired probabilistic recall guarantee.

In addition to evaluating the stopping rule under idealized conditions, this thesis investigates the robustness of the method to violations of its modeling assumptions. In real-world screening scenarios, the effectiveness of machine learning ranking may change during the screening process as relevant documents become more difficult to identify (Callaghan and Müller-Hansen, 2024a). To explore this, additional simulations are conducted in which the bias parameter governing ranking quality varies over time, allowing the potential impact of bias misspecification on the stopping rule to be studied in a controlled setting.

This thesis develops, evaluates, and tests a Bayesian stopping rule for machine learning-assisted systematic review screening. A computational implementation using vectorized algorithms is developed to enable efficient large-scale simulation experiments. The performance is then evaluated through Monte Carlo simulations across a range of experimental settings, before examining its robustness under scenarios where the effectiveness of the ranking model changes over the course of the screening process. Finally, the method is applied to real-world screening data from a collection of completed systematic reviews enabling a systematic comparison of the Bayesian stopping rule against a frequentist baseline across multiple reviews and confidence thresholds, evaluated on a high-performance computing cluster to enable analysis at scale. Together, these contributions provide both a theoretical validation and an empirical assessment of Bayesian stopping in realistic review conditions.

2 Literature Review

2.1 Machine Learning-Assisted Screening in Evidence Synthesis

The rapid growth of scientific literature has placed substantial strain on systematic review processes. Screening titles and abstracts to identify relevant studies is widely recognized as one of the most resource-intensive stages of evidence synthesis, with enormous resources currently devoted to screening records, critical evaluation, and data extraction (O'Mara-Eves et al., 2015).

Machine learning-assisted prioritization has emerged as a promising solution. Active learning approaches rank documents by predicted relevance, allowing reviewers to identify relevant studies earlier in the screening process. Numerous empirical studies demonstrate that prioritized screening can achieve high recall while substantially reducing workload (Yang et al., 2022).

However, while ranking methods have improved, a critical methodological bottleneck remains: determining when it is safe to stop screening. This stopping problem is the central unsolved challenge in practical application (Callaghan

et al., 2024). The absence of a trusted stopping criterion has concrete consequences for practice. The Cochrane Handbook for Systematic Reviews of Interventions — the authoritative methodological guide for evidence synthesis — does not currently recommend any stopping criterion for machine learning-assisted screening, precisely because no existing method can provide a true statistical guarantee that a sufficient proportion of relevant studies has been identified (Higgins et al., 2019). This thesis directly addresses that gap.

The potential for workload reduction through ML-assisted prioritization has been demonstrated across multiple domains. O'Mara-Eves et al. ((**?**)) find in a systematic review of text mining approaches that savings of between 30% and 70% of screening effort are potentially achievable, though the magnitude varies substantially with prevalence, corpus size, and the quality of the ranking model. This variability is itself an important finding: reviews with very effective ranking models realize the largest efficiency gains, while those with sparser evidence bases or weaker classifiers see more modest savings. Both the potential for efficiency gain and its dependence on ranking quality are directly relevant to the results presented in this thesis.

2.2 The Stopping Problem Under Uncertainty

In live review settings, recall is fundamentally unobservable until all documents have been screened (Callaghan and Müller-Hansen, 2020). Reviewers must therefore decide whether to stop while remaining uncertain about the total number of relevant studies. This creates a decision problem under uncertainty: stopping too early risks missing relevant evidence and introducing bias; stopping too late forfeits potential work savings. Crucially, the efficiency gains reported in existing studies may not translate to new reviews, as they often rely on retrospective data where the optimal stopping point is already known (Callaghan et al., 2024).

Several authors emphasize that existing systems lack robust stopping criteria, limiting their applicability in real-world review contexts. Simulation studies further confirm that screening performance varies substantially across review topics, prevalence levels, and model configurations, underscoring the difficulty of defining a universally safe stopping point (Ferdinands et al., 2023). In particular:

- Heuristic stopping rules (e.g., stopping after k consecutive irrelevant records) ignore sampling uncertainty and lack any formal statistical basis (Callaghan and Müller-Hansen, 2020).
- Sampling-based prevalence estimators often fail to correctly account for sampling uncertainty, producing errors that can have severe consequences (Callaghan and Müller-Hansen, 2020).
- Model-dependent stopping criteria may rely heavily on ranking performance and hyperparameter tuning, rendering them unreliable when ranking is imperfect (Repke et al., 2026).

Alternative approaches have sought to address these limitations through statistical modeling of the remaining relevant document population. Bron et al. (Bron et al., 2025) propose using Chao's estimator - a species richness estimator of ecology - as a stopping criterion. On the other hand, Stevenson and Bin-Hezam (Stevenson and Bin-Hezam, 2024) model the occurrence of relevant documents as a point process. Despite this diversity of approaches, no consensus has emerged on a universally reliable method.

Empirical evaluations confirm that almost all commonly used stopping rules are unsafe, either failing to achieve recall targets or requiring unrealistic assumptions (Repke et al., 2026). Heuristic methods that use the number of consecutive excludes tend to miss the majority of relevant records (Repke et al., 2026), and it has been argued in strong normative terms that arbitrary stopping criteria have no place in high-quality systematic reviews (Callaghan et al., 2024).

A recurring theme in the literature is that reliable stopping rules must: (1) explicitly quantify uncertainty; (2) allow specification of a target recall level; (3) provide interpretable confidence statements; and (4) remain robust to imperfect ranking performance (Callaghan et al., 2024). Despite nearly two decades of research in active learning for systematic reviews, early stopping remains controversial and is not universally recommended in review guidelines (Callaghan and Müller-Hansen, 2024a).

2.3 Hypergeometric Foundations of Statistical Stopping

A major methodological advance was the formulation of stopping as a hypothesis testing problem grounded in the hypergeometric distribution. When screening without replacement from a finite corpus of size N, the number of relevant documents observed after screening n records follows a hypergeometric distribution under random sampling. This finite-population model provides a principled basis for reasoning about the number of remaining relevant documents, and grounds stopping rules in finite-population inference rather than asymptotic binomial reasoning (Callaghan and Müller-Hansen, 2024b).

Stopping can be formulated as a hypothesis test where the null hypothesis asserts that the achieved recall is below a target threshold — equivalently, that the number of remaining relevant documents exceeds a critical value (Callaghan and Müller-Hansen, 2024b). Formally:

$$H_0 : \texttt{Recall} < \tau \tag{1}$$

where τ is a predefined recall target. Such methods allow reviewers to reject the null hypothesis with controlled Type I error, providing a formal statistical guarantee (Callaghan and Müller-Hansen, 2020).

However, prioritized screening violates the central hypergeometric assumption of random sampling. Since documents are ranked by predicted relevance, early draws are systematically biased toward relevant studies (Callaghan and Müller-Hansen, 2024a). Applying a central hypergeometric model in this context therefore leads to conservative stopping behavior: the test implicitly assumes a worst-case prior over remaining relevant documents, requiring more

evidence before rejecting H_0 than is strictly necessary (Callaghan and Müller-Hansen, 2024b). This over-conservatism has been empirically confirmed (Repke et al., 2026).

2.4 Biased Urn Models and Noncentral Hypergeometric Distributions

To address this mismatch, recent work has introduced biased urn formulations. Observing a long sequence of irrelevant documents intuitively suggests that the prevalence of remaining relevant documents is low, but the implication for recall depends jointly on the number of relevant documents already identified and the number of documents yet to be screened (Callaghan and Müller-Hansen, 2024a). Biased urn models make this reasoning precise.

Noncentral hypergeometric distributions generalize the central hypergeometric to account for differential selection probabilities (Fog, 2008a, 2008b):

- *Wallenius' noncentral hypergeometric distribution* models sequential biased sampling without replacement, where selection probabilities update as items are drawn.
- *Fisher's noncentral hypergeometric distribution* arises from conditioning independent biased draws on a fixed total sample size.

Both reduce to the standard hypergeometric when the odds ratio equals one (i.e., no bias between classes) (Fog, 2008a).

BUSCAR (Biased Urns for Stopping Criteria in Assisted Reviews) extends hypergeometric stopping by explicitly modeling the bias introduced by prioritized screening through an urn framework, rather than assuming random draws (Callaghan and Müller-Hansen, 2024a). Although BUSCAR is the only evaluated stopping method that never stops before a set recall target is achieved, empirical evidence indicates that its stopping rule remains conservative, particularly for high recall targets and low-prevalence settings, thus foregoing large shares of potential work savings (Repke et al., 2026). This conservatism is a direct consequence of the random-sampling assumption embedded in the underlying hypergeometric model (Callaghan and Müller-Hansen, 2024a).

A further limitation of the biased urn approach as currently implemented concerns the estimation of the bias parameter ω itself. In practice, ω is not directly observable and must be specified by the user prior to screening. Existing implementations have relied on rough heuristics or conservative default values, effectively amounting to an educated guess (Callaghan and Müller-Hansen, 2024a). This is problematic because the stopping behavior of the rule is sensitive to the assumed value of ω: an overestimated bias leads to overly optimistic recall estimates and premature stopping, while an underestimated bias produces unnecessary conservatism. A Bayesian framework naturally addresses this limitation by treating ω as an unknown parameter to be inferred jointly with K from the observed screening sequence, rather than fixed in advance.

2.5 Limitations of Existing Approaches

Across the literature, several limitations emerge:

- Heuristic rules lack statistical validity and misunderstand the significance of a low proportion of relevant documents in estimating recall (Callaghan and Müller-Hansen, 2020).
- Prevalence estimators based on point estimates ignore sampling uncertainty, offering poor reliability in terms of either recall or work saved (Callaghan and Müller-Hansen, 2020).
- Hypothesis-testing approaches impose implicit worst-case assumptions over remaining relevant documents (Callaghan and Müller-Hansen, 2024b).
- Stopping behavior remains overly conservative under bias-aware frequentist methods, limiting efficiency gains (Callaghan and Müller-Hansen, 2024a; Repke et al., 2026).
- Existing methods do not allow reviewers to specify a confidence level directly; they are not given the opportunity to define the probability with which they are willing to accept a given level of missed recall (Callaghan and Müller-Hansen, 2020).
- The bias parameter ω must currently be specified by the user in advance, based on a conservative estimation or assumption, rather than being inferred from the data (Callaghan and Müller-Hansen, 2024a).

Importantly, existing approaches largely frame stopping as a frequentist hypothesis test rather than as a posterior inference problem. Researchers developing stopping methods have explicitly called for uncertainty-aware methods that can estimate compounding effects in screening. Specifically, these are methods capable of quantifying the uncertainty that accumulates through both ranking imperfection and sampling variability (Repke et al., 2026).

2.6 Bayesian Approaches in Evidence Synthesis

Bayesian methods have been implemented in related areas of evidence synthesis, including meta-analysis (Gelman et al., 2013) and the modeling of publication bias, but their use in the stopping problem for prioritized screening remains largely unexplored. For instance, Bayesian hierarchical models are routinely used in meta-analysis to pool effect estimates across studies while accounting for between-study heterogeneity (Gelman et al., 2013), and Bayesian approaches have been proposed for the detection of publication bias in evidence synthesis pipelines. The closest existing work by Callaghan and Müller-Hansen (Callaghan and Müller-Hansen, 2020) introduces the idea of a posterior confidence threshold for recall but stops short of a fully Bayesian generative model of the screening process itself. Rather than constructing a hypothesis test about recall under a worst-case assumption about remaining relevant documents, the

framework in this thesis proposes to treat the unknown quantities, which are the total number of relevant documents K and the ranking bias ω, as parameters with prior distributions that are updated as each screening outcome is observed (Gelman et al., 2013). This produces a posterior distribution over recall at any given point in screening that supports direct probabilistic statements, selected confidence thresholds, and explicit quantification of uncertainty. These are features that frequentist stopping rules can only approximate indirectly(Berger, 1985).

2.7 Gap in the Literature

Although hypergeometric and biased urn models have been developed within a frequentist framework, a fully Bayesian stopping rule for prioritized screening that jointly infers both the total number of relevant documents and the ranking bias from observed data remains underexplored (Repke et al., 2026). The field has yet to reach a consensus on when it is safe to stop screening, and future work is explicitly needed to develop safe, robust stopping methods that can reliably save work by stopping early at a set confidence level (Repke et al., 2026). Specifically, the literature lacks posterior modeling of the total number of relevant documents under biased sampling; explicit credible statements about recall with user-specified confidence levels; and systematic empirical evaluation of posterior-based stopping across multiple review scenarios and prevalence settings. This thesis presents the first stopping method for systematic review screening to adopt a Bayesian framework, addressing the lack of calibrated uncertainty quantification and user-specified confidence levels that has been repeatedly identified as a shortcoming of existing approaches (Callaghan et al., 2024).

3 Methodology

3.1 Problem Setup

In systematic review screening, documents are ranked by a machine learning model that estimates the probability of relevance, and reviewers screen documents sequentially in descending order of predicted relevance (Cormack and Grossman, 2014). The objective is to develop a stopping rule that allows screening to terminate early while guaranteeing, with high probability, that recall exceeds a predefined target threshold.

Let N denote the total number of documents in the corpus and K the unknown total number of relevant documents. After screening n documents, let k_n denote the number of relevant documents identified. Recall at step n is then defined as:

$$R_n = \frac{k_n}{K} \tag{2}$$

Since K is unknown in a live review, R_n is unobservable in practice (Callaghan

and Müller-Hansen, 2020). Stopping therefore requires reasoning under uncertainty about K given the data observed so far.

3.2 Generative Model: The Biased Urn Process

The screening process is modeled as a biased urn process, in which documents are drawn sequentially without replacement from a corpus of N documents, of which K are relevant (Johnson and Kotz, 1977). Under unbiased (random) sampling, the number of relevant documents encountered after screening n records would follow a standard hypergeometric distribution (Callaghan and Müller-Hansen, 2024b). However, prioritized screening systematically draws relevant documents earlier, violating this assumption (Callaghan and Müller-Hansen, 2024a, 2024b).

To capture this bias, a Wallenius' noncentral hypergeometric distribution is adopted as the sampling model (Fog, 2008a). At each step, the probability of drawing a relevant document is:

$$p_{rel} = \frac{\omega \cdot R_{remaining}}{\omega \cdot R_{remaining} + I_{remaining}} \tag{3}$$

where $R_{remaining}$ and $I_{remaining}$ denote the number of remaining relevant and irrelevant documents respectively, and $\omega > 0$ is a bias parameter capturing the strength of the ranking model. When $\omega = 1$ the process reduces to random sampling; when $\omega > 1$ relevant documents are drawn earlier, consistent with effective ML-assisted prioritization. The parameter ω thus serves as a proxy for ranking quality, and its uncertainty is explicitly modeled alongside K.

3.3 Bayesian Inference: Posterior Updating

Rather than testing a null hypothesis about recall under worst-case assumptions, the stopping problem is reformulated here as a posterior inference problem. Specifically, the approach:

1. Places a prior over the unknown total number of relevant documents K and the bias parameter ω
2. Updates this prior sequentially using a likelihood that accounts for biased sampling
3. Computes the posterior probability of having achieved a target recall level
4. Allows the user to directly specify a confidence threshold γ at which screening is stopped.

This reframing replaces fixed significance tests with calibrated probabilistic statements about recall (Berger, 1985; Gelman et al., 2013).

The central inferential task is to characterize the joint posterior distribution over the unknown parameters K and ω given the observed screening sequence

$y_{1:n}$, where $y_i \in \{0, 1\}$ indicates whether the i-th document screened was irrelevant or relevant.

Following the principles of Bayesian updating (Gelman et al., 2013), the posterior after n steps is:

$$p(K, \omega \mid y_{1:n}) \propto p(y_n \mid K, \omega, y_{1:n-1})\, p(K, \omega \mid y_{1:n-1}) \tag{4}$$

As screening progresses, the posterior is updated sequentially: each new observation shifts probability mass toward values of K and ω consistent with the observed pattern of relevant and irrelevant documents (Gelman et al., 2013). This replaces the implicit worst-case assumptions of frequentist hypergeometric tests with an explicit, evolving belief distribution (Callaghan and Müller-Hansen, 2024b).

A uniform (non-informative) prior is placed over a discrete grid of values for $K \in \{k_n, k_n + 1, \ldots, N\}$ and $\omega \in \{\omega_{\min}, \ldots, \omega_{\max}\}$, consistent with standard practice for grid-based Bayesian inference (Gelman et al., 2013). The likelihood at each step is evaluated using Wallenius' noncentral hypergeometric probability mass function (Fog, 2008a). Crucially, by placing a prior over ω and updating it alongside K, the method infers the ranking bias directly from the observed screening sequence, rather than requiring the user to specify it in advance - directly addressing the limitation identified in existing biased urn approaches (Callaghan and Müller-Hansen, 2024a).

3.4 Stopping Rule

Let $\tau \in (0, 1]$ be a target recall level and $\gamma \in (0, 1]$ a desired confidence level. The Bayesian stopping rule stops the screening in the first step n such that:

$$\Pr(R_n \geq \tau \mid y_{1:n}) \geq \gamma \tag{5}$$

This posterior probability is computed by marginalizing over the joint posterior of K and ω:

$$\Pr(R_n \geq \tau \mid y_{1:n}) = \sum_{K,\omega} \mathbf{1}\left[\frac{k_n}{K} \geq \tau\right] p(K, \omega \mid y_{1:n}) \tag{6}$$

Stopping is triggered when this quantity first exceeds γ. Unlike frequentist stopping rules, which test a null hypothesis at a fixed significance level, this formulation directly answers the question of practical interest: with what probability has a given recall target already been achieved? (Berger, 1985). Both τ and γ are user-specified, enabling reviewers to calibrate the trade-off between recall assurance and workload reduction (Callaghan and Müller-Hansen, 2020). Moreover, Bayesian updating replaces the implicit worst-case prior of frequentist hypergeometric tests with a transparent, data-driven belief distribution that evolves as screening progresses. This provides more interpretable and flexible stopping behavior than existing approaches.

3.5 Computational Implementation

To make inference computationally feasible, the posterior is evaluated on a discrete grid of possible values for K and ω. The full posterior matrix is updated at each screening step using vectorized matrix operations in R, which allows simultaneous evaluation across all parameter combinations and avoids the computational cost of nested loops (Gelman et al., 2013). For computational tractability, K is evaluated on a coarse grid with step size 20, yielding approximately 500 grid points for a corpus of $N = 10{,}000$, and ω is evaluated on a grid of 15 uniformly spaced points between 1 and 6. This discretisation introduces a small approximation error relative to a full-resolution grid, but substantially reduces computational cost and enables large-scale Monte Carlo evaluation. The step size was chosen empirically to balance precision against runtime; sensitivity to grid resolution is acknowledged as a limitation in Section 5.6.

At each step, the likelihood vector over the grid is computed using Wallenius' noncentral hypergeometric probability mass function as implemented in the `BiasedUrn` package (Fog, 2008a). The posterior matrix is then renormalized and the stopping criterion evaluated. Without vectorization, the sequential updates required for live screening would become computationally infeasible for large scale datasets.

3.6 Monte Carlo Simulation Study

The performance of the stopping rule is evaluated using Monte Carlo simulations. For each experimental configuration, synthetic screening sequences are generated using the biased urn model, and the stopping rule is applied to each sequence. Performance is summarized across simulations using two metrics. Recall at stopping is defined as $R_{n_{stop}} = k_{n_{stop}}/K$, and work saved is defined as:

$$WorkSaved = 1 - \frac{n_{stop}}{N} \tag{7}$$

A stopping rule is considered safe if it achieves the target recall τ in all (or a specified high proportion of) simulations, and efficient if work saved is large (Repke et al., 2026). These two objectives are in tension: a more conservative rule is safer but saves less work, mirroring the core trade-off identified in the empirical literature (Callaghan and Müller-Hansen, 2024a).

3.7 Experimental Design

The experimental evaluation proceeds in three stages, each designed to assess a different aspect of the stopping rule's performance.

Stage 1: Controlled Simulation under Constant Bias. The baseline behavior of the stopping rule is first evaluated under controlled conditions where the model assumptions are satisfied. Synthetic screening sequences are generated using the biased urn model with four parameter configurations that vary

corpus size, prevalence, and ranking strength. The baseline configuration uses $N = 10{,}000$, a prevalence of 5% ($K = 500$), and $\omega = 3$. Sensitivity analyses additionally examine $\omega = 5$ to assess performance under stronger ranking, prevalence of 1% to simulate sparse corpora typical of large-scale evidence synthesis, and $N = 5{,}000$ to evaluate behavior on smaller datasets. Each configuration is repeated across 100 Monte Carlo runs, and the stopping rule is evaluated across nine combinations of $\tau \in \{0.90, 0.95, 0.99\}$ and $\gamma \in \{0.90, 0.95, 0.99\}$, yielding $4 \times 9 = 36$ experimental conditions in total. Performance is measured using recall at stopping and work saved as defined in Section 3.6.

Stage 2: Robustness under Time-Varying Bias. The second stage examines the behavior of the stopping rule when the constant bias assumption is violated. Four bias scenarios are implemented, as described in Section 5.2: constant bias (baseline), linear decay, sudden drop, and remaining relevance. All robustness experiments use $N = 10{,}000$, prevalence of 5%, $\tau = 0.95$, and $\gamma = 0.99$, with 100 Monte Carlo runs per scenario. The linear decay scenario is then extended to a sensitivity analysis over decay strength $\lambda \in \{0.25, 0.50, 0.75, 1.00\}$, and both the unadjusted and conservatively adjusted versions of the stopping rule are evaluated at each level, allowing the robustness benefit of the adjustment to be quantified as a function of violation severity.

Stage 3: Evaluation on Real-World Screening Data. The third stage evaluates the stopping rule on real-world systematic review screening data provided by Callaghan and Müller-Hansen via Zenodo. The dataset comprises screening outputs from ten completed systematic reviews, structured by review and run. For each of the ten reviews, 100 independent screening runs are available, yielding 1,000 run-level observations per confidence level and 3,000 in total across $\gamma \in \{0.90, 0.95, 0.99\}$. The Bayesian stopping rule and the frequentist baseline, specifically the hypergeometric stopping criterion of Callaghan and Müller-Hansen (2020), are applied to each run, and stopping points, achieved recall, and work saved are computed for both methods. Experiments were parallelized across runs and executed on a high-performance computing cluster to enable evaluation at scale. The reviews vary substantially in corpus size ($N = 258$ to $N = 8{,}793$), prevalence (0.3% to 8.2%), and effective ranking quality, providing a varied evaluation dataset for assessing the generalization of the simulation findings to realistic screening conditions.

4 Results

4.1 Controlled Simulation under Constant Bias

Results from the Stage 1 Monte Carlo simulation are summarized in Figures 1, 2, and 3, which show median recall at stopping, median work saved, and miss rate respectively across all four experimental configurations and nine τ/γ combinations.

4.1.1 Recall at Stopping

Across all configurations where the stopping rule triggered, median recall increases monotonically with both τ and γ, confirming that the method behaves as intended: higher recall targets and confidence levels produce more conservative stopping and therefore higher achieved recall (Figure 1).

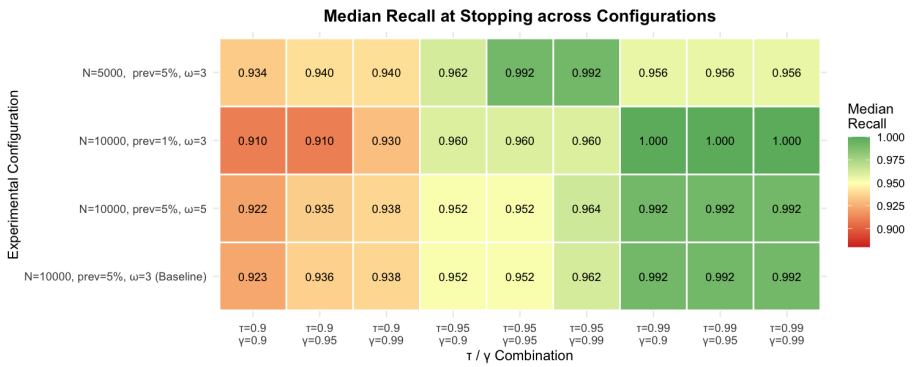

Figure 1: Median recall at stopping across four experimental configurations and nine combinations of recall target τ and confidence level γ. Each cell reports the median recall achieved across 100 Monte Carlo runs. Recall increases monotonically with both τ and γ across all configurations, confirming the method's intended behavior. The sparse prevalence configuration (N = 10,000, prevalence = 1%) achieves perfect median recall of 1.000 at $\tau = 0.99$, reflecting rapid posterior concentration when the number of relevant documents is small.

Under the baseline configuration (N = 10,000, prevalence = 5%, $\omega = 3$), median recall ranges from 0.923 at $\tau = 0.90$, $\gamma = 0.90$ to 0.992 at $\tau = 0.99$, confirming that the posterior correctly concentrates around the target level. The stronger ranking configuration ($\omega = 5$) produces qualitatively identical recall values, suggesting that recall at stopping is driven primarily by τ and γ rather than ranking strength. The sparse prevalence configuration (N = 10,000, prevalence = 1%) achieves perfect median recall of 1.000 at $\tau = 0.99$, reflecting the fact that with only 100 relevant documents in the corpus, the posterior becomes highly certain once the final relevant documents are identified.

4.1.2 Work Saved

Work saved decreases predictably as τ and γ increase, reflecting the fundamental trade-off between reliability and efficiency (Figure 2). Under the baseline configuration, median work saved ranges from 41.4% at $\tau = 0.90$, $\gamma = 0.90$ to 20.3% at $\tau = 0.99$. The stronger ranking configuration ($\omega = 5$) achieves substantially higher work savings across all parameter combinations, with median work saved of 57.7% at $\tau = 0.90$, $\gamma = 0.90$ and 37.6% at $\tau = 0.99$. This confirms that the method successfully exploits effective ML prioritization to stop earlier

when ranking quality is high, consistent with the biased urn generative model (Callaghan and Müller-Hansen, 2024a).

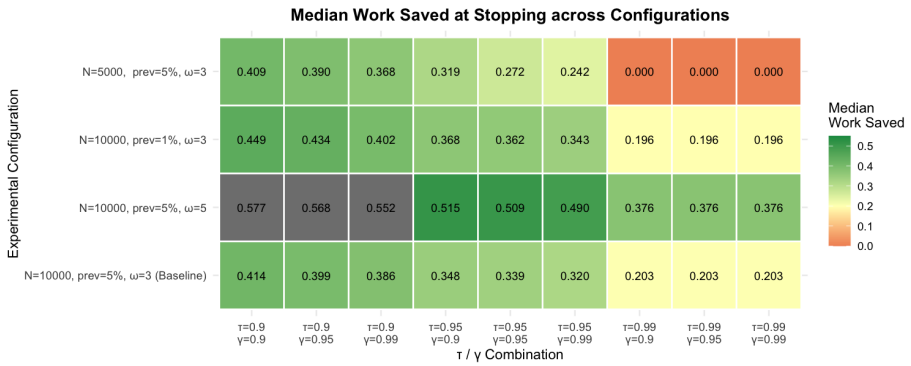

Figure 2: Median work saved at stopping across four experimental configurations and nine combinations of recall target τ and confidence level γ. Each cell reports the median proportion of documents not screened across 100 Monte Carlo runs. Work saved decreases as τ and γ increase, reflecting the fundamental trade-off between reliability and efficiency. The stronger ranking configuration ($\omega = 5$) achieves substantially higher work savings across all parameter combinations. Grey cells (value = 0.000) indicate configurations in which the stopping rule failed to trigger in any of the 100 runs, representing a boundary condition of the method.

The smaller corpus configuration ($N = 5{,}000$) reveals an important boundary condition: at $\tau = 0.99$, the stopping rule fails to trigger in any of the 100 Monte Carlo runs, producing zero work saved. This indicates that a corpus of 5,000 documents with 5% prevalence is too small for the posterior to achieve the required 99% confidence in having met the recall target, given the resolution of the parameter grid. This finding highlights a practical limitation of the method: in small reviews with high recall requirements, the posterior may never become sufficiently concentrated to trigger stopping, and reviewers would need to screen the full corpus.

4.1.3 Miss Rate and Guarantee Reliability

The miss rate heatmap (Figure 3) reveals that the reliability of the recall guarantee depends critically on the choice of γ. At $\gamma = 0.90$, miss rates are non-trivial across all configurations: under the baseline, 6% of runs miss the target at $\tau = 0.90$ and 8% miss it at $\tau = 0.95$.

These rates decrease substantially as γ increases, reaching zero for most configurations at $\gamma = 0.99$. The key practical implication is that $\gamma = 0.90$ provides insufficient protection against recall failures regardless of the recall target chosen, while $\gamma = 0.99$ produces reliable guarantees across all configurations

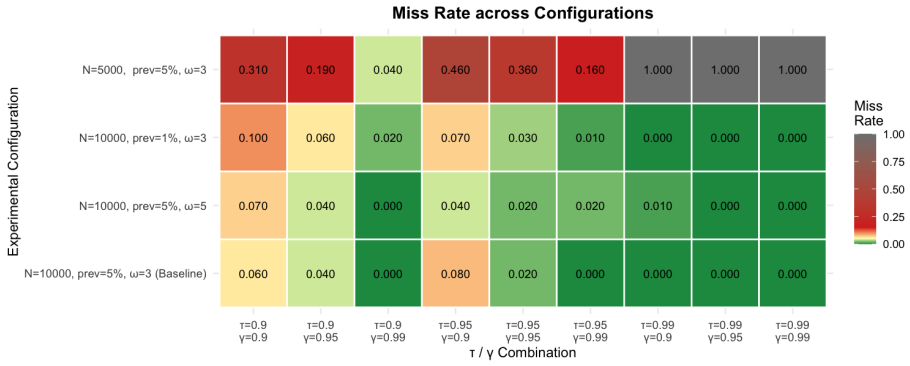

Figure 3: Miss rate across four experimental configurations and nine combinations of recall target τ and confidence level γ. Red cells indicate the proportion of runs in which the stopping rule triggered but achieved recall below the target τ. Grey cells (value = 1.000) indicate configurations in which the stopping rule failed to trigger in all 100 Monte Carlo runs, representing a complete failure of the method under those parameter settings. Miss rates decrease consistently as γ increases, approaching zero at $\gamma = 0.99$ for all large-corpus configurations. The $N = 5{,}000$ configuration (top row) shows substantially higher miss rates and complete stopping failure at $\tau = 0.99$, identifying a practical boundary condition of the proposed method.

where stopping triggers. The minimum practically reliable parameter combination therefore appears to be $\tau = 0.95$, $\gamma = 0.99$, which achieves zero miss rate in all large-corpus configurations. Under the baseline simulation configuration this combination saves a median of 32% of screening effort, though as the real-world evaluation in Section 4.2 demonstrates, work savings on actual review data are substantially lower, averaging 1.5%. This discrepancy reflects the difference between idealized simulation conditions and the diversity of ranking quality encountered in practice.

4.2 Performance on Real-World Screening Data

The Bayesian stopping rule and the frequentist baseline were evaluated on ten systematic reviews from the Callaghan dataset (Callaghan and Müller-Hansen, 2020), comprising 100 independent screening runs per review and three confidence levels ($\gamma \in \{0.90, 0.95, 0.99\}$), for a total of 3,000 evaluated runs. The results are summarized by confidence levels in Table 1 and by individual reviews in Table 2.

4.2.1 Reliability

The Bayesian stopping rule achieves lower miss rates than the baseline at every confidence level (Figure 4). At $\gamma = 0.90$, the Bayesian method misses the recall

target in 1.5% of runs, compared to 5.1% for the baseline. At $\gamma = 0.95$, miss rates are 0.9% and 2.3% respectively, and at $\gamma = 0.99$ they converge to 0.1% and 0.2%. The Bayesian method thus consistently provides a tighter probabilistic guarantee on recall across all confidence levels tested, confirming its intended behavior as a reliability-first stopping rule.

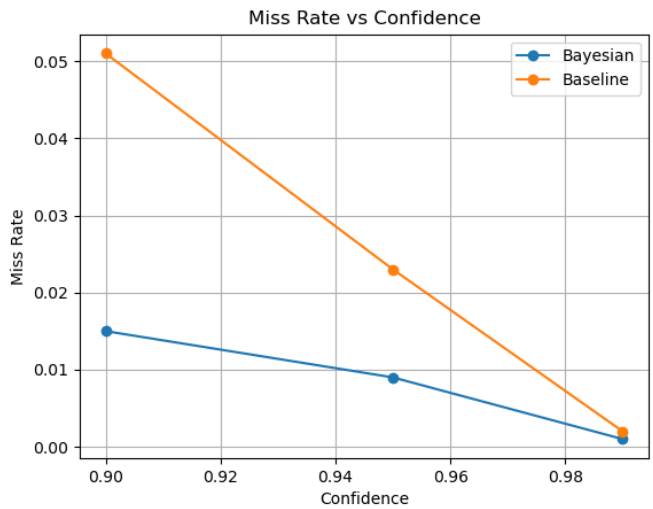

Figure 4: Miss rate as a function of confidence level (γ) for the Bayesian stopping rule and the frequentist baseline across all real-world reviews.

Table 1: Summary of miss rate and mean work saved by confidence level.

γ	Miss Rate (Bayesian)	Miss Rate (Baseline)	Work Saved (Bayesian)	Work Saved (Baseline)
0.90	0.015	0.051	0.019	0.208
0.95	0.009	0.023	0.015	0.176
0.99	0.001	0.002	0.010	0.120

4.2.2 Efficiency

The reliability advantage of the Bayesian method comes at a substantial efficiency cost (Figure 5). At $\gamma = 0.95$, the Bayesian method saves a mean of 1.5% of screening effort, compared to 17.6% for the baseline. This gap is consistent across all confidence levels and represents the primary trade-off of the proposed approach. Rather than a methodological shortcoming, this conservatism reflects the method's design: the Bayesian stopping rule maintains its guarantee regardless of ranking quality, while the baseline exploits high-quality rankings more aggressively.

Figure 5: Mean work saved as a function of confidence level (γ) for the Bayesian stopping rule and the frequentist baseline.

4.2.3 Review-Level Heterogeneity

The aggregate efficiency gap masks substantial variation across individual reviews. Figure 6 shows the distribution of additional effort required by the Bayesian method relative to the baseline at $\gamma = 0.95$, revealing a markedly bimodal pattern. Table 2 provides the full breakdown.

Table 2: Per-review statistics at $\gamma = 0.95$, sorted by prevalence.

Review	N	Prevalence	Work Saved (Bayesian)	Work Saved (Baseline)
8	5,812	0.003	0.003	0.043
4	1,630	0.006	0.005	0.010
3	1,908	0.008	0.003	0.015
6	8,793	0.012	0.024	0.510
12	5,214	0.021	0.023	0.034
10	882	0.042	0.009	0.095
5	258	0.058	0.000	0.019
11	975	0.076	0.018	0.254
9	7,216	0.081	0.037	0.498
7	1,175	0.082	0.026	0.284

Two reviews stand out as extreme cases. Review 6 (N = 8,793, prevalence = 1.2%) and Review 9 (N = 7,216, prevalence = 8.1%) see the baseline save approximately 51% and 50% of screening effort respectively, while the Bayesian method saves only 2.4% and 3.7%. These are also the two largest reviews in

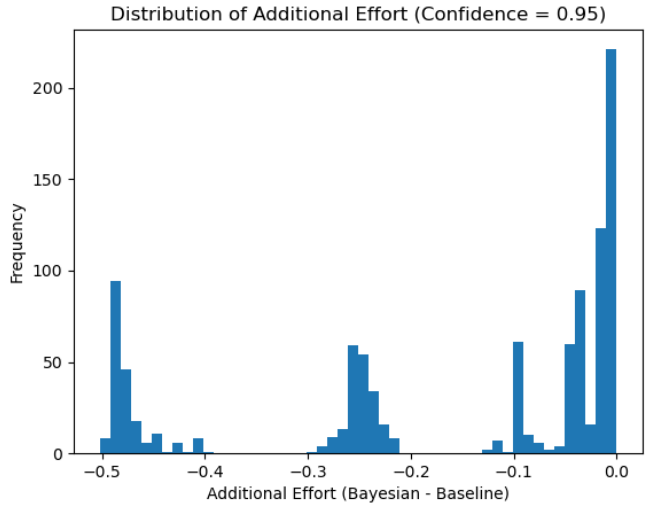

Figure 6: Distribution of additional screening effort required by the Bayesian method relative to the baseline ($\gamma = 0.95$). Negative values indicate that the Bayesian method screens more documents than the baseline.

the dataset, and both exhibit strong ML prioritization as evidenced by the baseline's ability to stop very early. This pattern reveals a key dynamic: the better the ranking model, the more aggressively the baseline exploits it, while the Bayesian method remains conservative regardless of ranking performance. In the majority of reviews, however, where the baseline saves modest amounts of work (1–10%), the absolute efficiency gap between the two methods is small. The bimodal distribution in Figure 6 reflects precisely this split between reviews with very effective and moderately effective ML prioritization.

4.3 Simulation Study: Bias Scenarios

Having established the method's performance on real-world data under naturally occurring screening conditions, the following sections examine its behavior when the key modeling assumption that the bias parameter ω remains constant throughout screening is explicitly violated. To evaluate the method under controlled conditions and examine its behavior when the constant bias assumption is violated, a Monte Carlo simulation study was conducted with $N = 10{,}000$, prevalence = 5% ($K = 500$), and 100 simulation runs per scenario. Four bias scenarios were implemented, with the recall target set to $\tau = 0.95$ and $\gamma = 0.99$.

The bias parameter ω was generated according to the following scenarios:

1. **Constant bias**: $\omega = 3$ throughout screening, corresponding to the model assumption.

2. **Linear decay**: ω decreases linearly from 5 to $5-\lambda\times 3$, where λ is a decay strength parameter, representing gradual deterioration of ranking quality.
3. **Sudden drop**: $\omega = 5$ for the first 30% of documents, dropping to $\omega = 2$ thereafter, simulating abrupt ranking deterioration.
4. **Remaining relevance**: $\omega = 1 + 4 \times (R_{remaining}/K)$, so that ranking strength declines as fewer relevant documents remain.

Results are summarized in Table 3. Under the constant bias scenario, the method behaves as expected: median recall at stopping is 0.972 and the 1% quantile of recall is 0.960, satisfying the target. Under linear decay, performance degrades modestly (1% recall = 0.924) but the method continues to stop relatively early (median work saved = 0.453). The sudden drop and remaining relevance scenarios produce the largest violations of the recall target, with 1% recall falling to 0.868 and 0.838 respectively. In these cases the inference model substantially overestimates ω late in screening, causing the posterior to become overly optimistic and triggering earlier stopping than warranted.

Table 3: Simulation results across bias scenarios ($\tau = 0.95$, $\gamma = 0.99$, $N = 10{,}000$, prevalence = 5%, $n = 100$ runs).

Scenario	Median Work Saved	Median Recall	1% Recall
Constant	0.286	0.972	0.960
Linear decay	0.453	0.954	0.924
Sudden drop	0.509	0.896	0.868
Remaining rel	0.398	0.876	0.838

4.4 Sensitivity to Decay Strength

To investigate how strongly the constant bias assumption must be violated before the recall guarantee breaks down, the linear decay scenario was extended by varying the decay strength parameter $\lambda \in \{0.25, 0.50, 0.75, 1.00\}$. Under this parameterization, ω decreases linearly from 5 to $5 - \lambda \times 3$ over the course of screening, so that $\lambda = 1$ corresponds to ω declining from 5 to 2 by the end of the review.

Figure 7 shows the 1% quantile of recall at stopping as a function of decay strength for the unadjusted method, with the red horizontal line marking the $\tau = 0.95$ recall target.

The results reveal a clear monotone relationship: as decay strength increases, the recall guarantee degrades. Notably, even at the mildest decay tested ($\lambda = 0.25$), the unadjusted method does not fully satisfy the guarantee, with the 1% recall falling to approximately 0.944. At maximum decay ($\lambda = 1.00$), the 1% recall drops further to approximately 0.924. This confirms that the stopping rule is sensitive to even moderate violations of the constant bias assumption, and that the degree of degradation scales predictably with the severity of the violation.

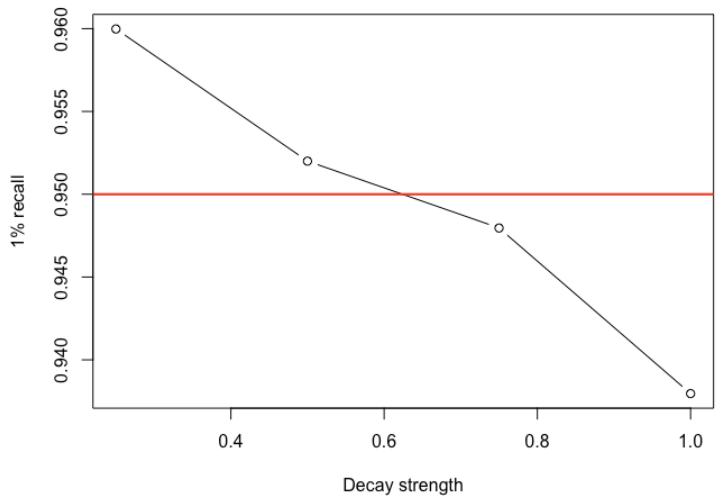

Figure 7: The 1% quantile of recall at stopping as a function of decay strength λ under the linear decay scenario without conservative adjustment. The red horizontal line marks the $\tau = 0.95$ recall target.

4.5 Conservative Adjustment

To mitigate the sensitivity to bias misspecification, a conservative adjustment was introduced in the Bayesian updating step. During inference, the effective bias parameter is down-weighted by a fixed factor of 0.8, so that $\tilde{\omega} = 0.8 \times \omega$ is used in the likelihood computation rather than ω directly. This causes the posterior to assign relatively more probability to lower values of ω, making the stopping criterion more conservative.

The same sensitivity analysis was repeated with the adjustment applied. Figure 8 shows both curves — unadjusted and adjusted — on the same axes. Several features of the plot are noteworthy. First, the conservative adjustment consistently raises the 1% recall across all decay strengths tested, shifting the blue curve upward by approximately 0.015–0.020 relative to the unadjusted black curve. Second, the two curves are approximately parallel, indicating that the adjustment provides a roughly constant improvement in recall guarantee regardless of the severity of the violation — it is a general correction rather than one effective only at a particular decay regime. Third, and most practically significant, the adjusted method satisfies the $\tau = 0.95$ guarantee up to approximately $\lambda = 0.75$, compared to no decay strength for the unadjusted method. Beyond $\lambda = 0.75$, even the adjusted method crosses below the recall target, with the 1% recall falling to approximately 0.942 at $\lambda = 1.00$.

These results demonstrate that simple, interpretable modifications to the Bayesian updating step can meaningfully improve robustness to assumption vi-

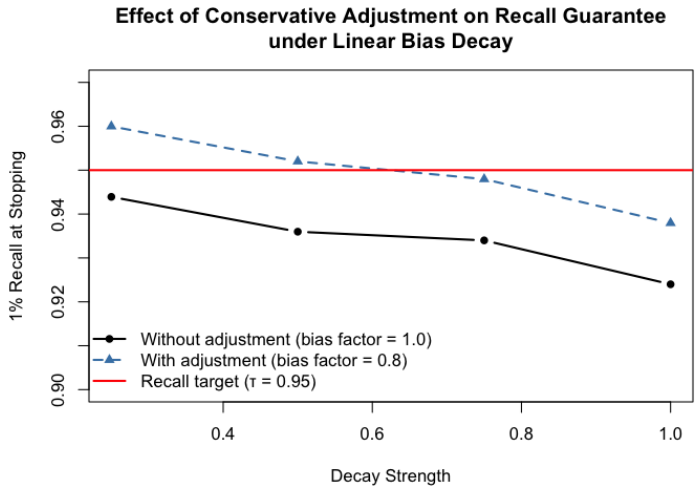

Figure 8: The 1% quantile of recall at stopping as a function of decay strength λ for the unadjusted Bayesian method (solid black) and the conservatively adjusted method with bias factor 0.8 (dashed blue). The red horizontal line marks the $\tau = 0.95$ recall target. The conservative adjustment extends the range of decay strengths under which the recall guarantee is maintained.

olations without requiring a fundamentally different model. The conservative adjustment introduces a deliberate asymmetry: by systematically underestimating ω during inference, the posterior remains uncertain about recall for longer before triggering the stopping condition. This is conceptually analogous to the conservative prior embedded in frequentist hypergeometric stopping rules (Callaghan & Müller-Hansen, 2020), but operationalized within the Bayesian framework in a transparent and tunable way. A natural direction for future work is to treat the bias factor as an additional unknown parameter to be inferred from the data, rather than fixing it at 0.8, which would allow the degree of conservatism to automatically adapt to the observed screening sequence (Gelman et al., 2013).

5 Discussion

5.1 A Conservative but Trustworthy Stopping Criterion

The controlled simulation results under constant bias confirm that the method behaves as theoretically expected: recall at stopping is driven by τ and γ, work savings scale with ranking quality, and the guarantee is reliably satisfied at $\gamma = 0.99$ across all large-corpus configurations. These findings provide the interpretive foundation for the real-world evaluation: the efficiency gap between

the Bayesian method and the baseline observed on the Callaghan dataset is consistent with the simulation results for moderate ranking strength ($\omega = 3$), while the reviews with very high work savings in the baseline (Reviews 6 and 9) correspond to the high-ranking scenario ($\omega = 5$) where the gap is largest.

The central finding is that the Bayesian stopping rule provides a more reliable recall guarantee than the frequentist baseline across all confidence levels and reviews tested, at the cost of substantially lower work savings. This trade-off is not incidental, as it reflects a deliberate design choice. The Bayesian method maintains its guarantee regardless of ranking quality, while the baseline exploits effective ML prioritization aggressively, stopping earlier when the ranking is strong but offering weaker recall assurance as a result (Callaghan and Müller-Hansen, 2024a; Repke et al., 2026).

In absolute terms, the work savings achieved by the Bayesian method are modest: a mean of 1.5% at $\gamma = 0.95$. However, in high-volume reviews this translates to a meaningful reduction in screening effort. In Review 6, with 8,793 documents, a 2.4% work saving corresponds to approximately 211 documents that do not need to be manually screened, while the method still provides a statistical guarantee that the recall target has been achieved. This guarantee is precisely what existing stopping rules cannot offer (Callaghan et al., 2024; Repke et al., 2026). In contexts where missing a relevant study has serious downstream consequences — including systematic reviews informing clinical guidelines, legal document review, and hate speech moderation — the reliability advantage of the Bayesian approach may outweigh the efficiency cost (Cormack and Grossman, 2014; Stevenson and Bin-Hezam, 2024).

This framing extends naturally beyond systematic review screening to any high-recall retrieval task where missing a relevant instance carries serious consequences. In legal discovery, for example, technology-assisted review is used to identify documents that must be disclosed in litigation, and failing to find relevant evidence can have severe legal consequences (Cormack and Grossman, 2014). Similarly, automated systems for identifying hate speech, detecting fraud, or flagging safety-critical incidents in large text corpora all share the same structure: a ranked candidate set must be screened until the reviewer can be confident that substantially all relevant instances have been found (Stevenson and Bin-Hezam, 2024). In each of these contexts, the question is not whether efficiency gains are possible, but whether those gains can be achieved with a genuine statistical guarantee. The Bayesian stopping rule developed in this thesis provides a principled, interpretable stopping criterion that trades some efficiency for a robustness guarantee that existing methods cannot offer.

5.2 The Reliability–Efficiency Trade-off Across Reviews

The review-level analysis reveals that the efficiency gap between the two methods is not uniform but concentrated in reviews where the ML ranking is highly effective. Reviews 6 and 9, both large corpora with strong prioritization, account for the most extreme divergence: the baseline saves approximately half of all screening effort in these reviews, while the Bayesian method saves only

2–4%. The bimodal distribution of additional effort (Figure 6) clearly shows that for the majority of reviews, the baseline offers only modest efficiency gains, and the absolute difference between the two methods is small.

This pattern has an important practical implication. In systematic review contexts where practitioners would most want a true statistical guarantee — precisely those contexts where screening is difficult, evidence is sparse, and missing relevant studies is costly — the ranking model is unlikely to be highly effective, and the efficiency cost of the Bayesian method will be correspondingly low. Conversely, in reviews where the ranking is very effective and the baseline saves large amounts of work, the stakes of missing a study may be lower, and the user might reasonably choose the baseline with its weaker guarantee. The choice between the two methods can therefore be informed by the expected ranking quality of the ML model for a given review.

5.3 Robustness to Bias Misspecification

The simulation study reveals that the Bayesian stopping rule is sensitive to violations of its constant bias assumption, but in a structured and predictable way. Under gradual linear decay of ω, the guarantee degrades monotonically with the severity of the violation. Even at the mildest decay tested ($\lambda = 0.25$), the unadjusted method falls slightly below the recall target, with the 1% recall at approximately 0.944. Strong decay ($\lambda \geq 0.60$) leads to more meaningful violations, with 1% recall falling to approximately 0.924 at $\lambda = 1.00$.

These findings align with the concern raised in the literature that misspecification of the bias parameter can lead to overly optimistic stopping (Callaghan and Müller-Hansen, 2024a). However, they also contextualize this concern: for moderate violations of the constant bias assumption — which are arguably more realistic than extreme sudden drops in ranking quality — the method remains approximately valid. The linear decay scenario with $\lambda \leq 0.30$ represents a situation where the ranking model gradually loses effectiveness as relevant documents become harder to find, which is a natural feature of many real-world screening tasks (Ferdinands et al., 2023).

Importantly, the sensitivity analysis serves a dual purpose. While it demonstrates that the method is sensitive to strong violations of the constant bias assumption, it also validates that assumption as a reasonable modeling choice. The alternative assumptions tested here — gradual decay, sudden deterioration, and dependence on remaining relevant documents — all produce worse outcomes than the constant bias model, both in terms of miss rate and stopping reliability. This suggests that assuming constant bias is not merely a convenient simplification, but a defensible and arguably conservative choice: by not attempting to model the dynamics of ω during screening, the inference procedure avoids overfitting to transient patterns in the ranking and maintains a more stable posterior. The advance over the frequentist status quo is therefore twofold: a less restrictive modeling assumption that produces a more conservative and trustworthy stopping criterion, at the cost of modest efficiency losses that are justified in contexts where the recall guarantee matters most.

5.4 The Conservative Adjustment

The conservative adjustment, implemented by down-weighting the effective bias parameter by a factor of 0.8 during inference, extends the range of assumption violations under which the recall guarantee is maintained. As shown in Figure 8, the adjusted method satisfies the $\tau = 0.95$ guarantee up to approximately $\lambda = 0.75$, compared to no decay strength for the unadjusted method, which falls below the target even at the mildest decay tested. This demonstrates that simple, interpretable modifications to the Bayesian updating step can meaningfully improve robustness without requiring a fundamentally different model.

With the conservative adjustment, the 1% recall remains above 0.95 up to approximately $\lambda = 0.75$, compared to no decay strength for the unadjusted method. The approximately parallel shift between the two curves across all decay strengths suggests that the adjustment acts as a general robustness correction rather than one tailored to a specific violation regime, making it a practically deployable safeguard for real-world screening settings where the true degree of bias decay is unknown in advance.

5.5 First Bayesian Stopping Rule for Systematic Review Screening

To the best of the author's knowledge, this thesis presents the first stopping criterion for machine learning-assisted systematic review screening to adopt a fully Bayesian framework. Existing approaches — including the hypergeometric stopping rule (Callaghan and Müller-Hansen, 2020), BUSCAR (Callaghan and Müller-Hansen, 2024a), and point process methods (Stevenson and Bin-Hezam, 2024) — are formulated as frequentist hypothesis tests or statistical estimators. While Bron et al. (Bron et al., 2025) and Stevenson and Bin-Hezam (Stevenson and Bin-Hezam, 2024) represent important recent advances in stopping methodology, neither adopts a Bayesian generative model that jointly infers both the total number of relevant documents and the ranking bias from the observed screening sequence.

The Bayesian framework offers three advantages over existing approaches that are demonstrated empirically in this thesis: lower miss rates at matched confidence levels; direct probabilistic statements about recall rather than hypothesis test outcomes; and a principled mechanism for introducing conservatism through prior or likelihood adjustment rather than through ad hoc parameter choices.

5.6 Limitations and Future Work

Several limitations of the current implementation should be acknowledged. First, the conservative adjustment factor of 0.8 was chosen heuristically and evaluated at a single value. A more principled approach would involve placing a prior over the bias factor itself, or learning it adaptively from the screening data — effectively extending the model to a hierarchical Bayesian framework (Gelman et al.,

2013). Second, the simulation study uses synthetic data generated under the biased urn model, which may not capture the full complexity of real-world ranking behavior. The PIK cluster evaluation on ten real reviews provides empirical validation, but a larger and more diverse set of reviews would strengthen the generalisability of the findings. Third, the method fails to trigger stopping in small corpora ($N \approx 5{,}000$) at high recall targets ($\tau = 0.99$), suggesting that the coarse parameter grid used for computational efficiency may be insufficient for the posterior to concentrate adequately in low-information settings. Future work should investigate adaptive grid refinement or importance sampling approaches to extend the method's applicability to smaller reviews (Gelman et al., 2013). Fourth, the current implementation assumes that the bias parameter ω is constant throughout screening and scalar. Extending the model to allow for time-varying or document-type-specific bias would address the sensitivity identified in the robustness analysis (Callaghan and Müller-Hansen, 2024a; Repke et al., 2026).

Future work might also investigate the integration of the Bayesian stopping rule into existing screening platforms such as ASReview or NACSOS (Repke et al., 2026), where it could be offered alongside the baseline as a high-guarantee alternative for users who prioritize recall assurance over efficiency. The correction factor approach developed here provides a practically deployable mechanism for this purpose, requiring only a single additional parameter and no changes to the underlying inference procedure.

6 Conclusion

This thesis set out to address a longstanding methodological gap in machine learning-assisted systematic review screening: the absence of a stopping criterion grounded in a fully Bayesian framework. Building on the biased urn model introduced by Callaghan and Müller-Hansen (Callaghan and Müller-Hansen, 2024a), a Bayesian stopping rule was developed that jointly infers the total number of relevant documents and the ranking bias parameter from the observed screening sequence, and stops screening once the posterior probability of having achieved a target recall level exceeds a user-specified confidence threshold.

Three contributions were made. First, a vectorized computational implementation was developed that makes large-scale Bayesian inference over a joint parameter grid tractable, enabling the method to be evaluated systematically across a wide range of experimental conditions. Second, the performance of the stopping rule was evaluated through Monte Carlo simulations across four configurations varying corpus size, prevalence, and ranking strength, demonstrating that the method reliably satisfies its recall guarantee at $\gamma = 0.99$ across all large-corpus settings, while saving between 20% and 58% of screening effort under idealized simulation conditions. On real-world screening data, mean work savings are more modest (1–2% at $\gamma = 0.95$), reflecting the diversity of ranking quality encountered in practice. Third, the method was applied to real-world screening data from ten completed systematic reviews, confirming that

it achieves consistently lower miss rates than the frequentist baseline at every confidence level tested, at the cost of substantially lower work savings.

The robustness analysis revealed that the method is sensitive to violations of its constant bias assumption, but in a structured and predictable way: mild violations preserve the guarantee, while strong violations lead to overly optimistic stopping. A conservative adjustment, implemented by down-weighting the effective bias parameter during inference, was shown to extend the range of assumption violations under which the guarantee is maintained, without requiring changes to the underlying model structure.

To the best of the author's knowledge, this thesis presents the first stopping criterion for systematic review screening to adopt a fully Bayesian framework. The results demonstrate that such a framework offers genuine advantages over existing frequentist approaches in terms of reliability and interpretability, while the conservative adjustment provides a practically deployable mechanism for improving robustness under realistic conditions. The fundamental trade-off between recall assurance and efficiency identified here — and the finding that this trade-off is most pronounced precisely in the reviews where ML ranking is most effective — represents an important empirical contribution to the growing literature on stopping criteria for technology-assisted review (Callaghan et al., 2024; Repke et al., 2026).

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